



Effectiveness of encoder-decoder deep learning approach for colorectal polyp segmentation in colonoscopy images

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Abstract

Colorectal cancer is considered one of the deadliest diseases, contributing to an alarming increase in annual deaths worldwide, with colorectal polyps recognized as precursors to this malignancy. Early and accurate detection of these polyps is crucial for reducing the mortality rate of colorectal cancer. However, the manual detection of polyps is a time-consuming process and requires the expertise of trained medical professionals. Moreover, it often misses polyps due to their varied size, color, and texture. Computer-aided diagnosis systems offer potential improvements, but they often struggle with precision in complex visual environments. This study presents an enhanced deep learning approach using encoder-decoder architecture for colorectal polyp segmentation to capture and utilize complex feature representations. Our approach introduces an enhanced dual attention mechanism, combining spatial and channel-wise attention to focus precisely on critical features. Channel-wise attention, implemented via an optimized Squeeze-and-Excitation (S&E) block, allows the network to capture comprehensive contextual information and interrelationships among different channels, ensuring a more refined feature selection process. The experimental results showed that the proposed model achieved a mean Intersection over Union (IoU) of 0.9054 and 0.9277, a dice coefficient of 0.9006 and 0.9128, a precision of 0.8985 and 0.9517, a recall of 0.9190 and 0.9094, and an accuracy of 0.9806 and 0.9907 on the Kvasir-SEG and CVC-ClinicDB datasets, respectively. Moreover, the proposed model outperforms the existing state-of-the-art resulting in improved patient outcomes with the potential to enhance the early detection of colorectal polyps.

Keywords Medical Image Segmentation · Semantic Segmentation · Polyp Segmentation · Deep Learning · Kvasir-SEG · CVC-ClinicDB

1 Introduction

Polyps are small abnormal growths on the colon or gastrointestinal tract that are precursors to colorectal cancer. These polyps can develop into Colorectal cancer if they are not diagnosed and treated at an early stage. Colorectal cancer ranks as the world's second most common type of cancer. Hence, colonoscopy plays a very critical role in accurately detecting and removing these cancerous polyps. Various studies reveal that 14% to 30% of polyps remain undetected even through colonoscopy due to polyps' size and texture. A study reported that colorectal cancer stands as the third major factor leading to death in the United States [1]. Therefore, it becomes crucial to diagnose and treat these polyps at an early stage to save patients' lives.

Normally, these polyps are detected through a process of colonoscopy, but it requires medical experts as well as enough time to thoroughly examine the data to detect

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polyps and other bleeding areas. Moreover, several factors like attention, expertise, and workload of medical experts can affect the manual examination of the polyps. Some other factors like size and color variations of polyps can also influence the detection and segmentation process [2]. Due to these reasons, polyps go undetected during colonoscopy with a miss-rate ranging from 6 to 27% [3]. Another problem is that polyps may look similar to other parts of organs. Therefore, polyps may remain un-detected if they are examined manually by doctors [4]. According to a recent survey, at least 25% of these polyps remain undetected if observed manually. In another recent study [5], it was found that although Computer-aided detection (CAD) systems help in finding the polyps which is very helpful but still the flat polyps [6] or the polyps smaller than 10 mm and sessile are still mostly missed during the examination process [7, 8].

However, the solutions proposed in existing studies have several limitations that hinder their practical implementation in the medical field. One of the primary issues is that the proposed models often wrongly segment some tissues as polyps leading to a decrease in the model efficiency. Some models also face the problem of overfitting as the size of training data is very small. To overcome those issues and effectively segment the polyps at an early stage, a CAD system with further improvements is needed. We aim to address the challenges associated with polyp segmentation. Our primary goal is to significantly enhance the accuracy of early polyp diagnosis during colonoscopy examinations, thus improving the overall diagnosing process to reduce the risk of polyps turning into colorectal cancer [6, 9]. We proposed a deep learning model that can efficiently segment the polyps, addresses the above-mentioned issues, and can be used as an early computer-aided automatic diagnosing system. Our contribution can be summarized as follows:

- By carefully configuring spatial attention and channel-wise attention mechanism along with other layers in the decoder block, we significantly improved the model performance.
- By optimizing the dual attention block and efficiently utilizing them in our model, we effectively capture global features and channel-wise interdependencies, leading to a marked improvement in overall segmentation performance.
- The proposed approach achieved state-of-the-art segmentation results on both publicly available datasets, Kvasir-Seg and CVC-ClinicDB.

The rest of the paper is organized into the following sections. Section 2 provides a brief overview of various methods used for the segmentation of colorectal polyps used in existing literature. Section 3 discusses the details of the steps performed in this study and provides details of the proposed

model. The experimental results are presented and discussed in Section 4. Finally, Section 5 concludes the findings of this study.

2 Literature review

In this section, we provide a concise overview of prior research related to polyp segmentation and medical image analysis. Recent advancements in computer vision, machine learning, and artificial intelligence have revolutionized almost all fields of life, especially the medical field. Artificial intelligence has played a vital role, particularly in the field of histopathologic image-based prediction. In the context of the COVID-19 pandemic, machine learning methods have played a pivotal role in predicting patient outcomes, modeling infection spread, and aiding in vaccine design [10, 11]. Another research [12] proposed a novel hybrid machine learning approach for predicting epitopes related to SARS-CoV-2 (Severe Acute Respiratory Syndrome Coronavirus 2), the virus responsible for COVID-19.

For the breast cancer diagnosis, Ahmed et al. [13] introduced a novel patch selection methodology to classify multi-resolution breast histopathological images. In their method, patches extracted from Whole Slide Images are fed into CNN extract features from those patches using the Dend-Clustering-based approach. Based on those features, the discriminative patches are fed into the Efficient-Net model. Efficient-Net then extracts features from those patches which are then used for classification of breast histopathological images. Extending their efforts in medical imaging diagnosis, they also proposed two methodologies [14, 15] specifically for segmenting various tissues and organs like liver, bone marrow, and spleen, for detailed body composition analysis using a 3-slice CT imaging protocol. The first approach utilized UNET++ based models while the second method employed regression analysis to perform voxel-wise statistical assessments. Their methods focus on segmenting complex anatomical structures that improve the visualization of critical areas within medical imaging.

Hayat & Aramvith [16, 17] introduced two methodologies for enhancing the resolution of medical imaging, specifically in the context of stereo endoscopic imaging. The first approach presented an edge-guided multi-stage network incorporating a Stereo Endoscopic Attention Module (SEAM), which progressively refines super-resolution from coarse to fine details. While in their subsequent work, they developed an innovative Saliency-Aware Super-Resolution deep learning model. They incorporate advanced cross-view feature interaction modules showcasing exceptional performance in enhancing image resolution, particularly at higher levels of scaling.

Extensive research has also been conducted particularly on colorectal polyp segmentation and several novel approaches have emerged to address the unique challenges within this field. Colorectal cancer is one of the most life-threatening cancers in the world. Early symptoms of this cancer include polyps and abnormal tissues growing inside the tract. Colon polyps are more likely to become cancer if left undetected and untreated [18]. Many computer-based models have been proposed to detect and segment those polyps. Most of the models follow encoder-decoder architecture. In recent years, UNet-based methods were mostly used for image segmentation and showed considerable performance. This section briefly overviews various methods used to segment colorectal polyps.

Tashk et al. [19] introduced a UNet model that is based on an encoder-decoder architecture with dilation convolution. The encoder captures multi-scale features from input data, whereas the decoder block maps those feature maps to segment the polyps. To capture high-level and important features, they used dilated convolutional layers in the encoder block without changing or reducing the resolution. They used various post-processing techniques including dropping and combining small objects, smoothening, and rescaling.

Inspired by the UNet [20] and FCN [21], Li et al. [2] introduced another Fully Convolutional Network model that comprises two stages named feature extraction and Prediction Map Reconstruction. The First stage captures the low-level features from the image and contains 8 Convolutional, 8 ReLu (Rectified Linear Unit), and 5 Pooling Layers. After going through these layers, the feature map is reduced to 1×1 and passed to the second stage. This stage changes this feature map back to the original image size using Deconvolutional, Concatenation, Convolutional, and ReLu layers, respectively. This model is robust and good enough to segment the polyps with similar colors as the background. This model achieved high accuracy on the CVC-ClinicDB dataset. However, the overall result demonstrated that this model doesn't segment wisely in the presence of reflecting light in the image.

Qiu et al. [22] proposed a BDG-Net (Boundary Distribution Guided) network that uses boundary information to extract the best features for segmentation. It consists of the BDGM (Boundary Distribution Generation Module), BDM (Boundary Distribution Module), and the BDGD (Boundary Distribution Guided Decoder). A BDGD block contains two other blocks called BDGD-A and BDGD-B at higher stages and lower stages, respectively. BDGD-A block receives two inputs from the previous decoder as well as its corresponding encoder. Whereas BDGD-B is similar to BDGD-A except that it doesn't have a branch to fuse encoders. Finally, the output of the BDGD-B block is convolved and passed through bilinear interpolation up-sampling to predict the

final mask. This achieved a mean IoU value of 86.5 on the Kvasir-SEG dataset.

Jia et al. [23] proposed a method named PLPNet which stands for 'Polyp Net'. They improved the FCN (Fully Convolutional Network) model by including polyp proposals at the region level, feature sharing, and skip schemes. PLPNet is divided into two parts. The first is the Polyp Proposal section, and the second is the Polyp Segmentation section. In the Polyp Proposal stage, the region-level polyp detection method is used which helps pixel-level learning whereas the second part is Polyp Segmentation. Which significantly improved the segmentation results.

Sang et al. [24] proposed a polyp segmentation model named AG-CUResNeSt. It consists of two coupled UNets having multiple encoders and decoders along with a skip connection between them and ResNeSt (Residual Neural Network with Split Attention) as the backbone. After feeding the input to the encoder block, it passes through five top-down blocks and down samples to a 16×16 feature map. The resultant feature map is passed through several decoder blocks and finally, when the output of the first UNet is generated, it is concatenated with the input image and passed to the second UNet. Resultant feature maps at various levels of the encoder of this second UNet are passed through the Attention gate and reach the relevant level of the decoder then the final output is generated.

Xiao et al. [25] used a pre-trained DeepLabv3 [26] as a backbone in the proposed model to extract the required features. They employed LSTMs (Long Short-Term Memory) in conjunction with DeepLabV3. For a long time, LSTMs have helped to manage finer details and other information related to extracted features. The pre-trained model also includes three additional blocks: ResNet, (Residual Network) the Multi-grid method, and ASPP (Atrous Spatial Pyramid Pooling). All of these blocks gather vital information that aids in the accurate prediction of polyps.

To address this issue of generalization, Shi et al. [27] introduced the SSFormer model which consists of a pyramid transformer encoder and a Progressive Locality Decoder (PLD). Inspired by the PVTv2 and SegFormer, they used ideas from both of these models and constructed the encoder named Pyramid Transformer Encoder.

Fang et al. [28] presented a model called ABC-Net (Area Boundary Constraint Network) that comprises a shared encoder and two mutually interactive decoders. The encoder used in this model is SE-ResNeXt-50 (Squeeze-and-Excitation ResNeXt) which is a modified version of ResNeXt-50. They used a series of S&E blocks in ResNeXt-50 to extract channel-wise features which increased the efficiency of the model. Two partially shared decoder blocks are used in this model. One for predicting the polyp area and the second one for the corresponding boundary or contours of that polyp.

To avoid the issue of loss of localized details and redundancy problems, Zhang et al. [29] proposed a segmentation model named TransFuse. It uses Convolutional Neural Network (CNN) and Transformers in a parallel way. Instead of using traditional fusion mechanisms, they used the BiFusion module which allows the fusion of multi-level features from both parts e.g., Transformers and CNN. By using this model, they were able to capture both fine-grained spatial details and overall global dependencies which considerably improved the model performance.

Fan et al. [30] proposed the PraNet (Parallel Reverse Attention) model to address the issue of polyps having unclear boundaries and varying sizes even with uncertain boundaries. The PraNet consists of three main components as an encoder which are Res2Net, PPD (Parallel Partial Decoder), and Reverse Attention (RA) module. PPD uses high-level features with parallel connections where the high-level feature maps are combined, and global maps are generated by utilizing these aggregated high-level feature maps which help the model in predicting the boundary of the polyp. They also used boundary cues by using RA blocks which allows them to understand the link between polyp areas and predicted boundaries with the help of recurrent cooperation mechanisms. This helps the PraNet to ignore some non-polyp areas that might be predicted as polyp otherwise.

Srivastava et al. [31] proposed a Multi-Scale Residual Fusion Network (MSRF) model that consists of four main components including Encoder Block, Stream block, MSRF sub-network, and Decoder block. In the encoder block, two convolution operations are performed on input and then passed through the S&E Block. This block improves channel interdependency and the computational cost for this process is much less. The squeeze step merges the feature maps whereas the Excitation step produces channel-wise dependencies. Finally, feature maps are down-sampled and passed to the MSRF subnetwork. As a result, accurate segmentation results can be produced while maintaining and improving information flow and propagating both high- and low-level characteristics.

After successful segmentation by MSRF Net, Srivastava et al. [32] proposed another model named GMSRF-Net (Global Multiscale Residual Fusion Network) which consists of an encoder, two GMSRF modules, and a decoder. They used CMSA (Cross Multi-Scale Attention) and GMSRF modules to incorporate better feature extraction mechanisms. They used Resnet50 as the backbone with ImageNet weights and incorporated the Receptive Field Block (RFB) to lower the computational cost. They used Vanilla Decoder where every block of decoder up samples the output of the previous decoder and concatenates it with the output of the GMSRF Module of the same scale.

Finally, the set of all output results from each decoder is transposed and the final segmentation mask is generated.

Thapa et al. [33] introduced a framework that combines uncertainty-based sampling and task-aware Coreset-based sampling. Integration of these two approaches improved the efficiency of sample selection for training the segmentation model. Uncertainty-based sampling helps identify the most uncertain or informative samples, allowing it to focus on challenging areas that require further annotation. Task-aware Coreset-based sampling leverages the concept of representative subset selection to choose diverse and informative samples that cover the entire data distribution. By combining these two sampling strategies, the overall accuracy and generalization of the polyp segmentation model are improved. This approach reduces the reliance on fully labelled datasets, thereby reducing annotation efforts and saving costs.

Yu et al. [34] proposed a new model called HarDNet-CPS (Harmonic Densely United Network for Colorectal Polyp Segmentation) for colorectal polyp segmentation that uses HarDNet-68 as the backbone. The HarDNet-CPS model improves local feature learning, facilitates small-scale object detection, and enhances spatial details in colonoscopy images. It also addresses the challenge of scale variation in polyps through MS-CAM (Multi-Scale Channel Attention Module) and ASFF. The proposed aggregation module further enhances feature expression and integrates cross-channel information for improved segmentation accuracy.

Fitzgerald et al. [35] introduced another model named FCB-SwinV2 (Fully Convolutional Branch Swin version 2) which is an updated version of the FCBFormer (Fully Convolutional Branch-Transformer) model. In this proposed model, the transformer block of FCBFormer is replaced with a SwinV2. The SwinV2 Transformer addresses training stability and resolution gaps by employing residual post-normalization and a scaled cosine function for attention computation. The SwinV2 Transformer network divides the input image into patches, applies linear embedding and patch merging layers, and employs successive SwinV2 transformer blocks with window-based multi-head self-attention. The FCB-SwinV2 Transformer incorporates a SwinV2 encoder and decoder blocks composed of SCSE modules, combining spatial squeeze and channel excitation for adaptive re-weighting of channel and spatial features. The FCB-SwinV2 Transformer system operates on 384×384 resolution images and includes changes inspired by the SwinV2 Transformer, such as increased channel dimensions and residual post-normalization in the FCB residual block.

Zhu et al. [36] proposed an encoder-decoder-based Global Context and Cross-Shaped Windows Vision Transformer Network (GCCSwin-UNet) that incorporates a patch-embedding layer, CSwin transformer block, and symmetric decoder. The patch-embedding layer is used for feature

down-sampling, and the symmetric decoder for feature up-sampling. CSwin Transformer block is used to extract contextual features. To improve the performance and enhance the feature learning capability of the model, a global context module (GCM) is utilized. The motivation behind this model was to address the limitation of CNNs in learning global and long-range semantic information.

Shen et al. [37] proposed a model that leverages HarD-Net68 as the backbone and integrates the attention mechanism in both the encoder and decoder parts. They employed a two-stage encoding decoding mechanism that helped in extracting global as well as local features, particularly for distinguishing polyps from normal mucosal tissues. This approach not only enables the model to minimize the loss during the encoding stage but also improves the information processing speed and computational cost.

Due to the limited size of medical imaging datasets and the associated challenges in achieving model generalizability, Ige et al. [38] proposed a model named ConvSegNet that utilized a novel CFR (Context Feature Refinement) module. Instead of using fixed kernel size, they used kernels with increasing size in multiple parallel convolutional layers. This approach significantly improved the generalization ability allowing it to extract finer details and identify the smaller polyps that might otherwise be unidentified.

Numerous models proposed for polyp segmentation primarily focused on improving accuracy while overlooking computational speed. To address speed issues, Tomar et al. [39] proposed an encoder decoder-based model named DilatedSegNet. It uses pre-trained ResNet50 as the backbone and incorporates several DCP (Dilated Convolution Pooling) blocks to process feature maps. This architecture achieved a remarkable real-time computation speed of 33.68 frames.

Sanderson et al. [40] proposed a model named FCB-Former that consists of two branches named FCN and Transformers branch. These branches work in parallel and extract crucial features from input. Feature maps produced by the Transformer branch are smaller in size ($h/4 \times w/4$) that is up sampled and concatenated with feature maps produced by the FCN branch. Finally, the Prediction Map block processes the final feature map and gives a full-size segmentation map.

Wen et al. [41] proposed another model that combines the strengths of FCN and CNN for mitigating the overfitting problem. They introduced a novel approach of weight transfer between dense (segmentation) and coarse (classification) vision. It is a two-stage architecture that involves generating pixel-wise regions followed by classification using binary CNN. ResNet50 is used as the backbone whereas another block for refining purposes is used to remove false-positive. This scheme effectively addressed the overfitting issue and improved segmentation accuracy.

Bhattacharya et al. [42] introduced the SMCA (Squeeze and Multi-Context Attention) module for better feature

extraction and representation. This attention module is designed to enhance the extraction of relevant features while suppressing irrelevant ones. In their approach, feature maps produced by the encoder block pass through the SMCA block before their transfer to the next encoding block. This block is used in the encoder block of several models and showed significant improvements in segmentation results.

Jha et al. [43] proposed an encoder decoder-based model named TransNetR that uses ResNet50 as the backbone. It uses a novel Residual Transformer block to process the feature maps generated by different levels of the encoder. This approach was tested on different unseen datasets and improved the real-time processing speed leading to improved generalizability.

To improve real-time computation speed and generalization, Tomar et al. [44] proposed a model called TransResUNet. This approach uses a transformer-based architecture and uses ResNet50 as the backbone. The output of the encoder block is passed through the transformer block and dilated convolution block and then concatenated. Which is then combined with the output generated by various levels of the encoder block and passed through the residual block. The output of the final residual block is convolved and passed through the sigmoid function to generate a segmentation mask.

Rahman et al. [45] introduced a model PVT-CASCADE for addressing transformers inability to learn local contextual dependencies among the pixels for better segmentation. They proposed a cascaded attention decoder to address the issue by incorporating multiple transformers in the hierarchy. The attention mechanism is used to improve long-range and local features among pixels by suppressing irrelevant features. Furthermore, they implemented a multi-stage loss optimizer that optimizes various losses from different hierarchical decoder stages resulting in faster convergence.

Zhang et al. [46] address limitations in CNN and Transformer models in capturing long-range dependencies with high computational efficiency. They propose a hybrid model combining the State Space Model (SSM) with Transformers in a U-shaped architecture called HTM-UNet, leveraging Mamba layers to efficiently model long-range spatial dependencies. Comprehensive experiments conducted on multiple public datasets, including ISIC17, ISIC18, and Kvasir, indicate that HTM-UNet demonstrates competitive performance in segmentation tasks, establishing a basis for further exploration of SSMs in medical imaging.

Fan et al. [47] address the limitations of Mamba-based segmentation models in capturing dependencies between neighboring features, which are crucial for accurately segmenting medical images. They propose SliceMamba, which includes a Bidirectional Slice Scan (BSS) module for bidirectional feature slicing, maintaining spatial coherence in the scanning sequence. To accommodate diverse

lesion and organ shapes, an Adaptive Slice Search method is employed to select optimal feature slices. Experiments on skin lesion, polyp, and multi-organ datasets, including ISIC2017, ISIC2018, and Synapse, validate SliceMamba's effectiveness, highlighting the significance of its adaptive slicing mechanism in enhancing segmentation performance.

Further, Kang et al. [48] introduce MCA-SAM, a Multi-scale Contrastive Adaptor learning method for enhancing the Segment Anything Model (SAM) in tasks with limited data. The model includes two adaptors: a Token-level Contrastive (TC) adaptor for local refinement and a Sample-level Contrastive (SC) adaptor for global understanding. Empirical results show substantial performance improvements, such as a 20.0% MAE increase on COD10K and a 7.9% mDice improvement on the Kvasir-SEG dataset.

Patel et al. [49] propose a dense decoder architecture designed to address the limitations of vision Transformers in capturing local relations across multi-level layers. They introduced the Dense Attention Gate (DAG) module, which aggregates multi-level dense features hierarchically within a nested decoder, thereby improving semantic feature representation. Evaluated on multiple polyp datasets, including Kvasir and ClinicDB, the model achieved state-of-the-art performance, underscoring the effectiveness of DAG in enhancing local feature relations across encoder layers.

Zhang et al. [50] extend the application of Mamba models with VM-UNetV2, a State Space Model-based UNet model designed to capture long-range dependencies with linear computational efficiency. The model incorporates Visual State Space (VSS) blocks and a Semantics and Detail Infusion (SDI) mechanism, enhancing feature fusion at various levels. Pre-trained on Mamba weights, VM-UNetV2 demonstrated competitive performance in segmentation tasks.

Wu et al. [51] introduced H-vmunet, a novel model based on the state-space model (SSM) that integrates High-order 2D-selective-scan (H-SS2D) techniques for medical image segmentation. The H-SS2D component significantly enhances global receptive fields while minimizing redundant information, with the Local-SS2D module improving local feature learning across different interaction levels. By reducing parameters by 67.28% compared to traditional Vision Mamba UNet models, H-vmunet demonstrated competitive performance across multiple datasets, including ISIC2017, Spleen, and CVC-ClinicDB.

Huang et al. [52] proposed YOLO-Med, an efficient end-to-end multi-task network designed to concurrently perform object detection and semantic segmentation in biomedical image analysis. This model includes a backbone with a neck structure for multi-scale feature extraction and incorporates two specialized decoders for task-specific processing. Additionally, they designed a cross-scale interaction module to enable effective information exchange between tasks, which

improves the model's performance and speed across biomedical datasets, including Kavarsir-seg.

The study by Biswas et al. [53] presents a deep learning-based approach to segment various organs including liver, lung, and polyps. They introduced a deep learning-based model that leverages a momentum term within residual blocks, enhancing training dynamics for medical image segmentation and classification tasks.

Nam et al. [54] presented M3FPolypSegNet, a frequency-based fully convolutional neural network. The model decomposes input images into low, high, and full-frequency components, utilizing three independent encoders to map these into high-dimensional feature spaces. Their Frequency-ASPP Scalable Attention Module (F-ASPP SAM) applies attention across different frequency scales, ensuring crucial spatial details are preserved while reducing information loss in up-sampling stages. Through multi-task learning across region, edge, and distance attributes, M3FPolypSegNet outperformed existing models on the CVC-ClinicDB and BKAI-IGH-NeoPolyp datasets, achieving state-of-the-art accuracy.

Lan et al. [55] developed BRAU-Net++, a hybrid CNN-Transformer architecture that combines the local feature extraction strength of CNNs with the global feature capture capabilities of transformers, thereby balancing complexity and computational efficiency. Utilizing bi-level routing attention within a u-shaped encoder-decoder structure, the network efficiently learns both local and global semantic information. Moreover, the Skip Connection with Channel-Spatial Attention (SCCSA) enhances spatial detail retention while allowing robust cross-dimensional interactions.

Jiang et al. [56] developed LV-UNet using pre-trained MobileNetv3-Large models and integrates fusible modules to reduce computational complexity and model size, making it suitable for deployment in real-time medical devices. It enables a switch to deployment mode at inference, lowering parameter counts and computation demands.

Tomar et al. [57] introduced FANetv2, a novel encoder-decoder model that incorporates a feedback attention mechanism that iteratively refines segmentation masks, with an initial mask generated by Otsu thresholding. Additionally, the model integrates text-guided attention, informed by polyp attributes such as quantity and size, to enhance feature representation. The summary of existing studies is provided in Table 1.

3 Research methodology

This section provides a detailed overview of the steps performed and how we conducted our experiments and discusses implementation details, hyperparameters, datasets, and the preprocessing techniques that we used. This section

Table 1 Summary of previously proposed models

Model	Architecture	Aim	Strengths	Weakness
UNet with dilation [19]	UNet-based with dilation convolutions; encoder-decoder; dilated convolutional layers for multi-scale features	Enhanced feature extraction using dilation convolutions	Precise polyp segmentation, and efficient feature extraction	May struggle with complex polyp backgrounds
FCN-based [21]	FCN with two stages; deep feature extraction and map reconstruction	Robust segmentation for polyps with similar colors as the background	Deep feature learning, and accurate map reconstruction	High computational costs for finer predictions and large filter sizes hinder learning
BDG-Net [22]	Uses EcientNet-B5 as backbone and introduced to novel module BDM and BDGD for boundary refinement	Accurate polyp segmentation using boundary guidance	Fast inference speed, very low computational cost	
PLPNet [23]	Extended fully convolutional ResNet-50 with a region proposal network (RPN) and Feature Pyramid Network (FPN)	Improved segmentation results using region-level detection	Robust region-based detection, high accuracy	It uses a deeper network which can lead to degradation problems, longer training times, and increased risk of overfitting
AG-CUResNeSt [24]	AG-CUResNeSt with coupled UNets, ResNeSt backbone, and attention gates	Improve feature fusion and attention for accurate segmentation	Effective multi-scale feature aggregation, better generalization	Complex model and the use of attention gate drop precision score with higher computational cost
DeepLabv3 with LSTM [25]	Combines DeepLab_v3 (with ResNet, multi-grid methods, and ASPP) and Long Short-Term Memory (LSTM) networks in parallel	Enhancing polyp detection by preserving polyp location information and improving feature extraction	Handling challenging textures, at a faster speed with 0.023 s per image	Computationally intensive, more parameters due to LSTM,
Mask R-CNN [58]	ResNet-50 as backbones and Region Proposal Network for selecting feature maps	Accurate and efficient polyp segmentation using ensemble methods	High-speed segmentation, and accurate localization	Lack of generalizability, less robust
SSFormer [27]	Utilized Swin Transformer, with a lightweight multilayer perceptron (MLP) decoder	Enhance semantic segmentation performance with reduced transformer-based model size and complexity	Suitable for low-resource computers, Enhanced feature extraction, and target-aware segmentation	More computational complexity, a high number of parameters
ABC-Net [28]	Uses SE-ResNeXt-50 encoder and dual decoders for polyp and boundary prediction Utilizes S&E Blocks for feature extraction	To improve the accuracy of colorectal polyp scanning	Accurate polyp boundary localization	Difficulty in locating smaller polyps, relatively poorer segmentation for some polyps
ColonFormer [59]	Utilizes a hierarchical Transformer & pyramid CNN, and Residual Axial Attention module for feature fusion	To improve long-range and multi-level feature representation	Efficient and accurate polyp segmentation, improved boundary refinement	computationally intensive and time-consuming
DS-TransUNet [60]	Dual Swin Transformer U-Net (DS-TransUNet) with hierarchical Swin Transformer in the encoder and Transformer Interactive Fusion (TIF)	To enhance semantic segmentation quality for medical images	Effectively models coarse and fine-grained features,	Potential computational complexity
FANet [61]	Incorporates SE-Residual blocks for feature extraction. —Implements MixPool blocks for hard attention	To enhance image segmentation by utilizing information from different training epochs	Efficient memory usage, requires less training time	Higher inference time

Table 1 (continued)

Model	Architecture	Aim	Strengths	Weakness
PRAPNet [62]	Utilizes ResNet50 as backbone; attention blocks; conditional random field	To improve intestinal polyp segmentation using global contextual information	Effective utilization of global contextual information	Additional computation due to two-branch architecture, resource-intensive
TGANet [63]	Text-guided attention with ResNet50 encoder, label attention, and multi-scale feature aggregation for adaptive polyp segmentation	Improved segmentation for differently sized polyps using text-guided attention	High mean IOU value Improved adaptability to polyp size variation, generalizes well to different datasets	Miss polyps
TransNetR [43]	Transformer-based Residual Network with ResNet50 encoder;	Improved real-time processing speed, particularly for out-of-distribution medical images	Achieves high performance on both in-distribution and out-of-distribution datasets, real-time processing speed,	Lack of robustness, less generalize to the unseen dataset
TransResU-Net [44]	Used ResNet-50 as the backbone, used transformer self-attention mechanism, dilated convolutions	To improve real-time processing speed	Improved real-time processing speed, simple lightweight architecture	Less generalize on the unseen dataset
DilatedSegNet [39]	pre-trained ResNet50 as backbone, Dilated Convolution Pooling Module for processing feature maps	To improve real-time processing speed	Less computational cost, Achieved real-time operation speed of 33.68 FPS	May struggle when adapting to multi-center datasets due to variations in data distribution
ConvSegNet [38]	Used a novel Context Feature Refinement Module, used kernel with increasing size in parallel convolutions	To improve model generalizability for out-of-distribution dataset	Shows better Generalization and FPS speed	Lower processing speed, and higher complexity than several other models
SMCA Block [42]	Uses Squeeze and excitation block, use average and max pooling layer	To improve mode accuracy and processing speed	Better generalizability for seen dataset, improved boundary refinement	Model do not generalize well on unseen dataset
PVT CASCADE [45]	Uses PVTv2 Encoder as the backbone, incorporates a novel attention-based decoder using a hierarchy of transformers	To address transformers inability to learn local contextual dependencies among the pixels	Faster convergence, improved generalization, and learning ability	Use multiple transformers in a hierarchy that may be cost-expensive
HMT-UNet [46]	UNet-based architecture with hybrid Mamba and Transformer	Leverage SSM and Transformer for enhanced long-range feature modeling	Efficiently models long-range dependencies, competitive segmentation performance	As it uses both the SSM and transformer, it may be computationally expensive
SliceMamba [47]	Mamba-based model with Bidirectional Slice Scan (BSS) module; utilizes NAS for optimal feature slicing	Improve segmentation by modeling both local and long-range features	Effective local feature modeling, strong performance on varied medical datasets	Not Reported
MCA-SAM [48]	SAM with Multi-scale Contrastive Adaptors (MC-adaptors), combining token-level and sample-level contrastive adaptors	Enhance SAM adaptability for polyp segmentation and other specialized tasks	Effective feature contrast across samples and patches, strong performance on diverse benchmarks	May struggle in camouflage scenarios requiring extensive fine-tuning or larger datasets
Multi-layer Dense Decoder [49]	PVT encoder with dense multi-layered decoder incorporating Dense Attention Gate and Camouflage Attention Module	To improve local relation learning across layers and Enhance feature aggregation in the decoder	Improved boundary delineation, effective multi-scale feature aggregation	Use of multiple stacked decoding layers may increase model complexity

Table 1 (continued)

Model	Architecture	Aim	Strengths	Weakness
VM-UNetV2 [50]	SSM-based UNet using Vision State Space (VSS) blocks with a Semantics and Detail Infusion (SDI) module	Capture long-range dependencies using SSM-based method and reduce the computational cost	Efficient computation due to the use of VSS blocks, Enhanced feature fusion of low-level and high-level details	Not Reported
H-vmunet [51]	SSM-based unet model using high-order visual state space (H-VSS) module	To improve medical image segmentation by reducing redundant information and enhancing local feature learning	Achieves a global receptive field with minimized redundant information, reduces parameter count by 67.28% compared to the traditional Vision Mamba UNet	Not Reported
YOLO-Med [52]	Shared encoder with CSPDarknet53 backbone and FPN neck, two task-specific decoders, and a cross-scale task-interaction module	To perform object detection and semantic segmentation simultaneously for biomedical image analysis	Efficient in both object detection and segmentation	Not Reported
Momentum-PVTFormer [53]	Residual network integrated with a momentum-based residual block	Improve Segmentation of organs from CT and MRI scans and classification of diseases	Improved convergence and stability in training, Effective at preserving memory due to reversible residual blocks	Not Reported
BRAU-Net+ + [55]	U-shaped hybrid CNN-Transformer network incorporating bi-level routing attention and channel-spatial attention	To achieve better segmentation by utilizing the merits of both CNN and transformer	Reduced computational complexity, Effective learning of local-global semantic information	Large increase in parameters for a slight improvement in dice value
LV-UNet [56]	Lightweight and vanilla model with pre-trained MobileNetV3-Large model	Design a compact and deployable model suitable for real-time medical image segmentation	Reduced both parameters count and computational load, efficient performance on multiple datasets, real-time application capability	Not Reported
FANetv2 [57]	Enhanced encoder-decoder with Pyramid Vision Transformer (PVT) and feedback attention mechanism	Improve segmentation process in different lightening conditions and rare cases	Improved feature representation, real-time high-processing speed of 36.64	High number of parameters

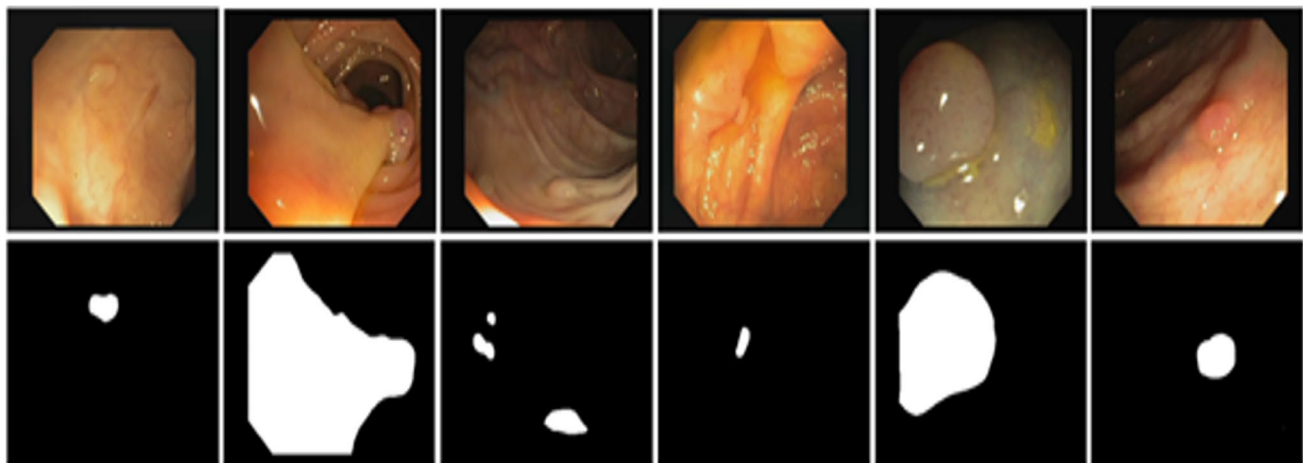


Fig. 3 The first row shows some example images from the CVC-ClinicDB dataset, and the second row shows their corresponding masks

Table 2 Datasets used in our experiment

Dataset	Images	Training	Validation	Split Ratio	Image Size
Kvasir-SEG	1000	700	300	70:30	Variable
CVC-ClinicDB	612	428	184	70:30	384 X 384

training. Since the size of the datasets used in this study is relatively small, which can lead to overfitting. Therefore, it becomes helpful to manually increase the size of the dataset by applying data augmentation. As the size of the Kvasir-SEG and CVC-ClinicDB datasets is very small, direct training of these datasets can cause overfitting of the model. There are various augmentation techniques such as crop, rotation, vertical or horizontal flip, and translation. In the proposed methodology, we applied horizontal flip and image rotation of 30, 60, 90, 120, and 150 degrees to all images. The results of applying augmentation techniques to

one sample image from each dataset are shown with the help of the following figures.

Figure 4 shows the result of applying augmentation techniques to an image from the Kvasir-SEG dataset. The first image in the first row is the original image whereas each next image is rotated 30 degrees clockwise. The second row shows the result of applying a horizontal flip to all images after applying rotations.

Figure 5 shows the result of applying augmentation techniques to an image from the CVC-ClinicDB dataset. The first image in the first row is the original image whereas each next image is rotated 30 degrees clockwise. The second row shows the result of applying a horizontal flip to all images after applying rotations.

3.3 Proposed network architecture

The model comprises multiple encoder and decoder blocks, with an intermediate Resblock (Residual Block) between the

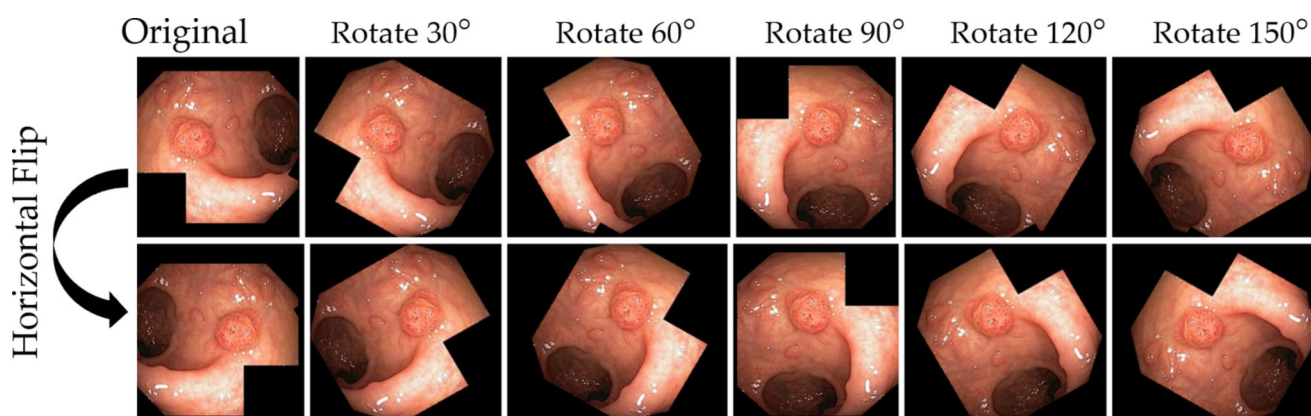


Fig. 4 Result of applying augmentation to a sample image from the Kvasir-SEG dataset

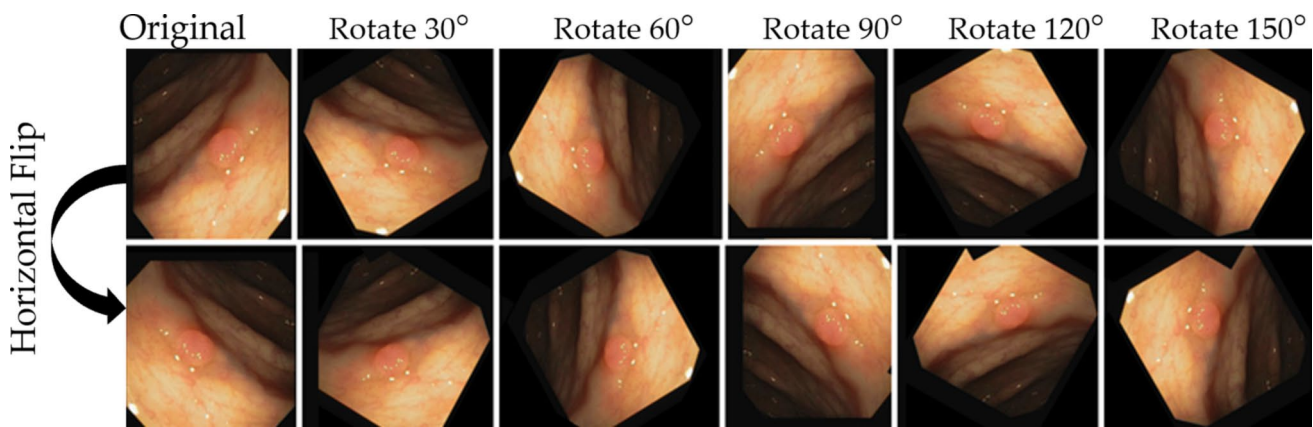


Fig. 5 Result of applying augmentation to a sample image from the CVC-ClinicDB dataset

last encoder and the first decoder block, resulting in a total of 39.9 million (39,955,195) trainable parameters. In our proposed model, normalized images with size 256×256 are fed into the model. These are passed through multiple encoder blocks getting down-sampled, the Resblock, and finally the decoder blocks. A dropout layer is also used after the last and second last decoder blocks. Feature maps are extracted by using a convolutional layer. Which is then down-sampled and passed to the next encoder block as well as to its corresponding decoder block through skip connections. The proposed network architecture is demonstrated in Fig. 6.

3.3.1 Encoder

The encoder block, as shown in Fig. 7(a), is one of the most important components of our model. It is designed to progressively capture complex features. Each encoder block contains two sets of convolutional layers, batch normalization, and ReLU activation, followed by a max-pooling

layer. The max-pooling operation, with a pool size of 2×2 , reduces the input dimensions by half while doubling the number of filters, allowing the model to capture more detailed features at each depth. After the first ReLU activation in the block, the output is convolved again, enhancing feature representation. The feature maps generated at each encoder stage are both passed forward to the next encoder block and linked via skip connections to the corresponding decoder block. These skip connections help maintain high-resolution details, accelerating convergence and improving model accuracy.

3.3.2 Decoder

The decoder block comprises an Attention block, a dual attention block, and a UP (Upsampling) block as shown in Fig. 7(d). UP block is followed by two sets of convolutional layers, batch normalization, and the ReLU activation function, ensuring smooth feature refinement at

Fig. 6 Proposed model architecture

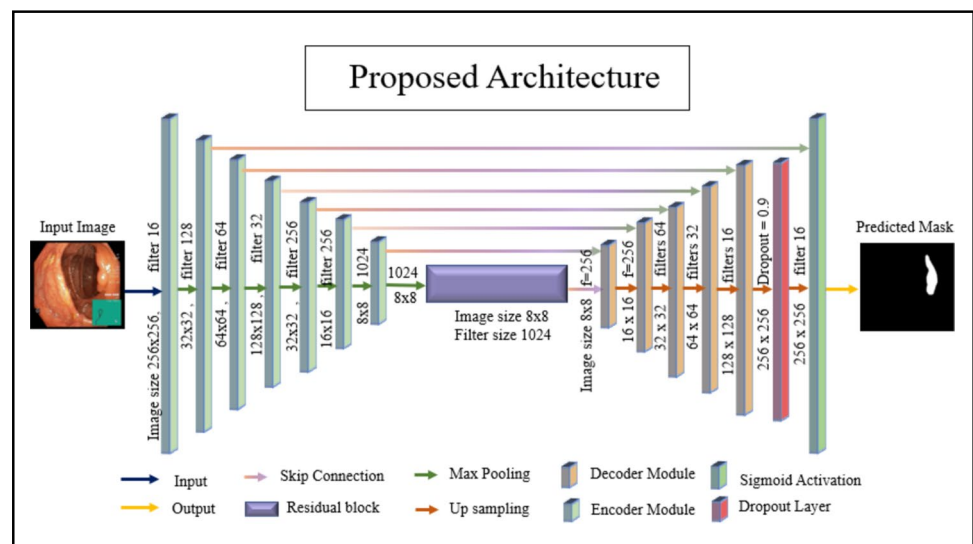
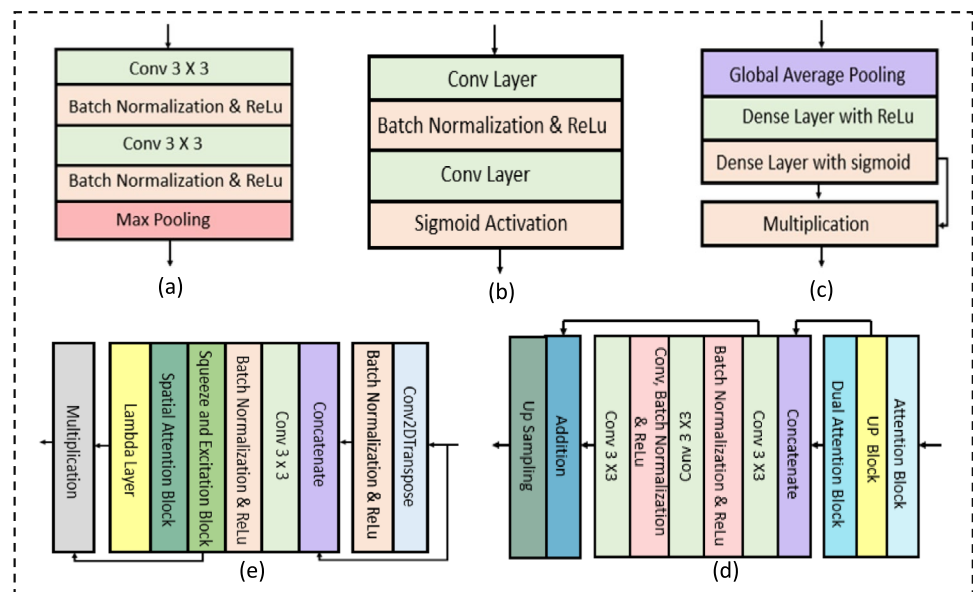


Fig. 7 (a) Encoder Block (b): Spatial Attention Block (c): Squeeze and Excitation Block (e): Dual Attention Block (d): Decoder Block



each stage. After passing through this ReLu function, the output is again convolved and added to the output of the first convolutional layer in this block. We used channel-wise and spatial attention modules in the decoder block to highlight the important features as well as interdependencies between each channel. Squeeze and Excitation Block is used to find channel-wise interdependencies whereas the spatial attention module highlights the important regions.

Each decoder block except the first block receives input from its corresponding encoder through skip connection and its previous decoder block. The first decoder block receives inputs from the last encoder block as well result of max pooling on this encoder's output. The decoder block consists of three blocks named attention block, UP block, and dual attention block sequentially. A dual attention block is followed by two sets of convolution layers with kernel and stride sizes of 3 and 1 respectively, batch normalization and ReLu activation. Output generated by these two sets is passed through another convolution layer which is then added to the concatenated output of both the UP block and attention block. The result of the convolution operation on this result is finally up-sampled using a stride of 2 and 'bilinear' interpolation.

A drop-out layer with a value of 0.9 is used after the last decoder module. After passing through the last decoder block followed by a dropout layer, a convolutional layer with a single output channel and the sigmoid function are applied to generate the final output.

3.3.3 Dual attention block

The attention mechanism is very important because it helps in focusing on the required target area in the image and improves model generalization. The dual Attention mechanism is also used in the proposed method. The dual attention block used in this model can extract important features from the input for a better representation of feature maps. It can extract features of interest from input having low lighting or visibility. This block uses two attention mechanisms named S&E and spatial attention block which are discussed next. The dual attention block is shown in Fig. 7(e).

3.3.4 S&E block

In our model, channel-wise attention is implemented through the Squeeze-and-Excitation (S&E) block, which is designed to focus on the most informative channels by learning inter-channel dependencies. This mechanism selectively enhances important feature channels and suppresses less relevant ones, improving the model's ability to capture global context. This block operates by first applying global average pooling to the input feature map, generating a global descriptor for each channel. These global descriptors are then passed through two fully connected layers. The first layer reduces the channel dimension (a bottleneck operation), while the second restores it. A sigmoid activation is applied to produce attention weights for each channel, which are multiplied element-wise with the original feature map to

scale the channels based on their importance. This way only those feature maps are preferred which are meaningful and have a better impact on the output. This block is used for extracting global features as well as for finding channel-wise interdependency between each channel. This also helps in suppressing those features which are irrelevant. Squeeze and excitation block is shown in Fig. 7(c).

3.3.5 Spatial block

After applying channel-wise attention, we employ a spatial attention block to refine the spatial regions of interest. This block first applies a convolutional layer to reduce the input channels to one. This is followed by batch normalization and a ReLU activation. Another convolution layer is used to generate the spatial attention map, which is then normalized using a sigmoid activation function. The resulting attention map highlights important spatial regions, helping the model to focus on critical parts of the image, such as the polyp regions. This spatial attention is multiplied element-wise with the input feature map, emphasizing regions of interest for improved segmentation results. The spatial attention block is shown in Fig. 7(b).

3.3.6 Activation function

There are several activation functions but the most commonly used are Sigmoid, ReLU, and Tanh. In our proposed model, sigmoid and Relu activation functions are used.

The sigmoid function, also known as the logistic activation function is the most commonly used. The curve of this function is S-shaped. Input to this function is real values whereas values between 0 and 1 are produced as output because the numerator is 1 and the denominator is always greater than 1. The following equations represent the sigmoid function:

$$\text{sigmoid}(x) = \frac{1}{1 + e^{-x}} \quad (1)$$

Here ‘e’ is a mathematical constant named Euler’s number which is equivalent to 2.71828.

The second activation function we used is called a Rectified Linear Unit which can receive positive and negative input values. This function returns the same output as input if the value received is a positive number (greater than zero). Otherwise, it returns to zero as output.

$$\text{Relu}(x) = \max(x, 0) \quad (2)$$

Here ‘max’ is a function that returns the maximum of both inputs e.g., ‘x’ and ‘0’. Therefore, if the value of x

is positive (greater than zero), its output will be the same positive number otherwise it returns to zero. Whereas in backward propagation, the equation of ReLU is as follows:

$$\text{Relu}(x)' = \begin{cases} 1, x > 0 \\ 0, x \leq 0 \end{cases} \quad (3)$$

There are two other activation functions derived from this function namely LeakyRelu and ELU (Exponential Linear Unit) both of which are helpful when the gradient value is less than zero.

3.4 Evaluation metrics & other hyperparameters

We used several evaluation metrics to evaluate the performance of the model including mean intersection over union (mean IoU), Dice Coefficient, Precision, and Recall, and compare the results of the proposed model with other state-of-the-art models. The following equations are used to calculate the value of mean IoU, Dice Coefficient, Precision, Recall, and Dice Loss [66].

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (4)$$

$$\text{Dice} = \frac{2|S \cap G|}{|S| + |G|} \quad (5)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (6)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (7)$$

$$\text{DiceLoss} = 1 - \text{Dice} \quad (8)$$

Here TP, TN, FP, and FN are True Positive, True Negative, False Positive, and False Negative respectively. Whereas ‘S’ represents the segmentation result or predicted output and ‘G’ represents the ground truth image. Whereas $|S \cap G|$ represents the overlapping area between the ground truth image and the predicted image. Hyperparameter tuning was conducted through a hit-and-trial approach by experimenting with various combinations of learning rates, batch sizes, and

Table 3 Hyperparameters

Batch Size	Optimizer	Learning Rate	Loss Function
8	Nadam	0.0015	Dice

optimizers. The hyperparameters used in this study are given in Table 3.

3.5 Implementation Detail

The proposed model was implemented using Python version 3.10.12, TensorFlow version 2.17.0, and Keras 3.4.1. Training and evaluation were carried out on a machine with the CPU Intel (R) Core (TM) i5-4590 processor, 16 GB of RAM, and a clock speed of 3.30 GHz. The GPU was an NVIDIA GeForce GTX 1050 Ti and had 12 GB/s memory bandwidth. The operating system was Microsoft Windows 10, and the software used was the Anaconda with Jupyter platform, CUDA Toolkit 8.0. By utilizing this hardware and software setup, we trained and evaluated the proposed model.

4 Results & discussion

To demonstrate the effectiveness of the proposed model, we validated our model on both datasets Kvasir-SEG and CVC-ClinicDB, and results show that it achieves better performance than various state-of-the-art models on both datasets. Although the size of both datasets is very small, which may impact the performance of various models. However, our model maintains high performance and shows robustness and good ability to various data distributions. Unlike other models that are based on boundary information or region-level detection, our model performs pixel-wise training on CNN, enabling more detailed and accurate segmentation results.

Moreover, utilizing the Spatial Attention (SA) mechanism, Channel-wise Attention (CA) mechanism and S&E blocks sets our model apart from other state-of-the-art models. SA and CA mechanisms capture the relevant features from important regions and suppress irrelevant information. The use of the S&E block further improves performance by capturing global features. It also finds channel-wise interdependencies among various channels and highlights the best features that improve the overall performance. Results on both datasets are shown in the tables as well as the graphs, highlighting the superior performance and advantages of our approach.

Results on the Kvasir-SEG dataset We trained our model on the Kvasir-SEG dataset and compared results with other SOTA models. Results show that the proposed model outperformed various SOTA (state-of-the-art) models with a mean IoU of 90.54, Dice coefficient of 90.06, Recall of 91.90, Precision of 89.85, and Accuracy of 98.06. The following graphs show the results of our model on the Kvasir-SEG

dataset. Table 4 shows the comparison of our proposed model with existing SOTA polyp segmentation models on Kvasir-SEG datasets. Results show that our model achieved the highest mean IoU on the Kvasir-SEG dataset.

Figure 8 shows a comparison of segmentation masks predicted by the proposed model and the ground truth mask. The figure clearly demonstrates that our model successfully segments polyps, especially flat ones or those with very low contrast and boundaries similar to the background, across different lighting conditions. However, there are still some instances where our model struggles to segment certain polyps effectively. This includes cases where images contain tissue types resembling polyps or have multiple small polyps in a single image as can be seen in Fig. 9. To illustrate the performance of the model for the KvasirSEG dataset, we present training and validation metrics using graphical representations, as seen in Fig. 10.

Results on CVC-ClinicDB We also trained our model on the CVC-ClinicDB dataset and compared results with other SOTA models. Results show that the proposed model also outperformed various SOTA models on the CVC-ClinicDB dataset with a mean IoU of 92.77, Dice coefficient of 91.28, Recall of 90.94, Precision of 95.17, and Accuracy of 99.07.

Table 5 shows the comparison of our proposed model with state-of-the-art polyp segmentation models on the CVC-ClinicDB dataset.

The results shown in Table 5 demonstrate that our model achieved the highest mean IoU on the CVC-ClinicDB dataset. We also show the results predicted by the model in Fig. 11 below. It shows a comparison of the actual polyp image, its corresponding ground truth mask, and the mask predicted by our model.

Figure 11 clearly shows that our model successfully segments polyps, especially flat ones or those that appear similar to the background, across different lighting conditions. However, there are still some cases where our model had difficulty in segmenting certain polyps effectively. This includes cases where images contain tissue types resembling polyps or have multiple small polyps in a single image as can be seen in Fig. 12. To illustrate the performance of the model for the CVC-ClinicDB dataset, we present training and validation metrics using graphical representations, as seen in Fig. 13.

The proposed model shows exceptional performance and has significant implications both in theory as well as practice. It achieves state-of-the-art results in polyp segmentation that was previously difficult with traditional segmentation models, particularly for flat and small polyps. Hence the proposed model is a major contribution in the field of medical image analysis for early detection of colorectal

Table 4 Comparison of Proposed Model with state-of-the-art models on the Kvasir-SEG Dataset

Sr. #	Model	Mean IoU	Dice	Recall	Precision
1	ColonFormer-S [65]	0.875	0.924	-	-
2	Polyp-PVT [67]	0.864	0.917	-	-
3	PlutoNet [68]	0.808	0.894	-	-
4	PRAPNet [62]	-	0.828	-	-
5	HSNet [69]	0.877	0.926	-	-
6	LAPFormer-L [70]	0.866	0.917	-	-
7	SAU-Net [71]	0.838	0.851	-	-
8	Li-SegPNet [72]	0.880	0.905	-	-
9	DS-TransUNet-L [60]	0.859	0.913	0.936	0.916
10	MSRF-Net [31]	0.891	0.921	0.919	0.966
11	GMSRF-Net [32]	0.884	0.926	0.940	0.931
12	FANet [61]	0.815	-	0.815	0.815
13	TGANet [63]	0.833	0.898	0.913	0.912
14	TA-Coreset [33]	0.892	-	-	-
15	GCCSwin-UNet [36]	0.831	0.863	-	-
16	FCB-SwinV2 [35]	0.897	0.942	0.934	0.958
17	FCBFormer [40]	0.879	0.890	-	-
18	TransNetR [43]	0.801	0.870	0.884	0.907
19	TransResU-Net [44]	0.821	0.888	0.910	0.902
20	DilatedSegNet [73]	0.833	0.895	-	-
21	HarDNet-CPS [34]	0.856	0.911	-	-
22	HarDNet with attention & multi-scale coding module [37]	0.862	0.915	-	-
23	Squeeze and Multi Context Attention Module [42]	0.780	0.850	0.890	0.860
24	PVT-Cascaded Attention Decoder [45]	0.877	0.925	-	-
25	ConvSegNet [38]	-	0.861	0.9124	0.869
26	HMT-Unet [46]	0.856	0.922	-	-
27	Momentum-PVTFormer [53]	0.852	0.907	0.931	0.917
28	MCA-SAM (ViT-H) [48]	0.845	0.917	-	-
29	SliceMamba [47]	0.873	0.932	-	-
30	Multi-Layer Dense Attention Decoder [49]	0.869	0.919	-	-
31	VMUnetV2 [50]	0.841	0.913	-	-
32	YOLO-Med [52]	0.886	-	-	-
33	LV-UNet [56]	0.8069	0.8917	-	-
34	Proposed Model	0.905	0.900	0.919	0.898

polyps. Achieving state-of-the-art results using optimized configuration and implementation of spatial and channel-wise attention mechanisms encourages researchers to further explore additional improvements in deep learning models and several other blocks that have not been fully explored. By significantly improving the overall segmentation process, especially flat and small polyps, our research sets a new benchmark for the theoretical advancement of medical image analysis.

The proposed model also offers significant implications for medical professionals and practitioners. With high mean IoU, precision, and recall rates, our model significantly reduces the polyp miss rate making it effective for

utilization in the medical field. This ensures that the medical practitioners get more accurate results, reducing unnecessary follow-ups and minimizing the risk of missing polyps. In addition, the proposed model performs effectively under varying lighting conditions and extracts important features, making it a state-of-the-art tool in the real-world medical field. The model also significantly decreases false-positive predictions thereby reducing the need for unnecessary medical procedures. This improvement can potentially lead to cost savings in healthcare. Additionally, it can serve as a valuable tool for real-world medical applications, offering precise polyp segmentation and potentially reducing the risk of missing polyps ultimately improving patient survival rate.

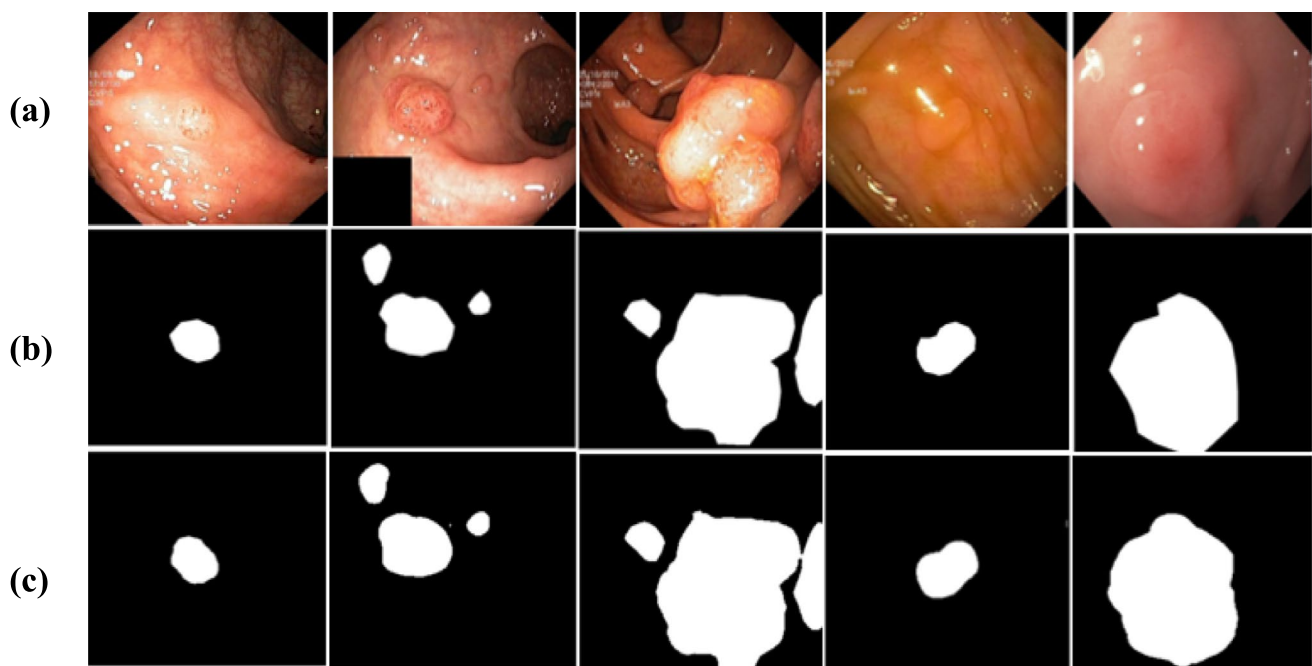


Fig. 8 Qualitative Results of Proposed Model on Kvasir-SEG dataset, (a): original image, (b): ground truth, (c): predicted mask

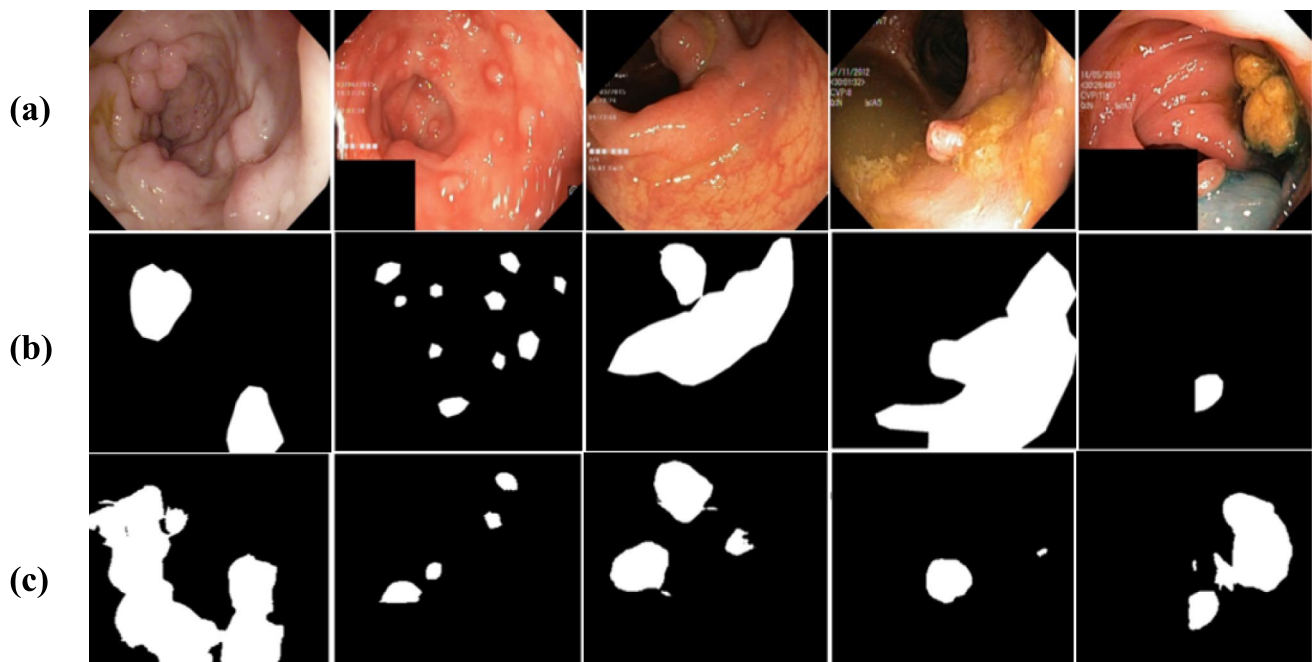


Fig. 9 Examples of challenging cases in KvasirSEG dataset, (a): Original Image, (b): Ground Truth, (c): Predicted Mask

5 Conclusions

This study presents a deep learning model utilizing encoder-decoder architecture for the efficient

segmentation of colorectal polyps in colonoscopy images. The model incorporates S&E blocks and spatial attention modules, which enhance its ability to focus on critical features while suppressing irrelevant

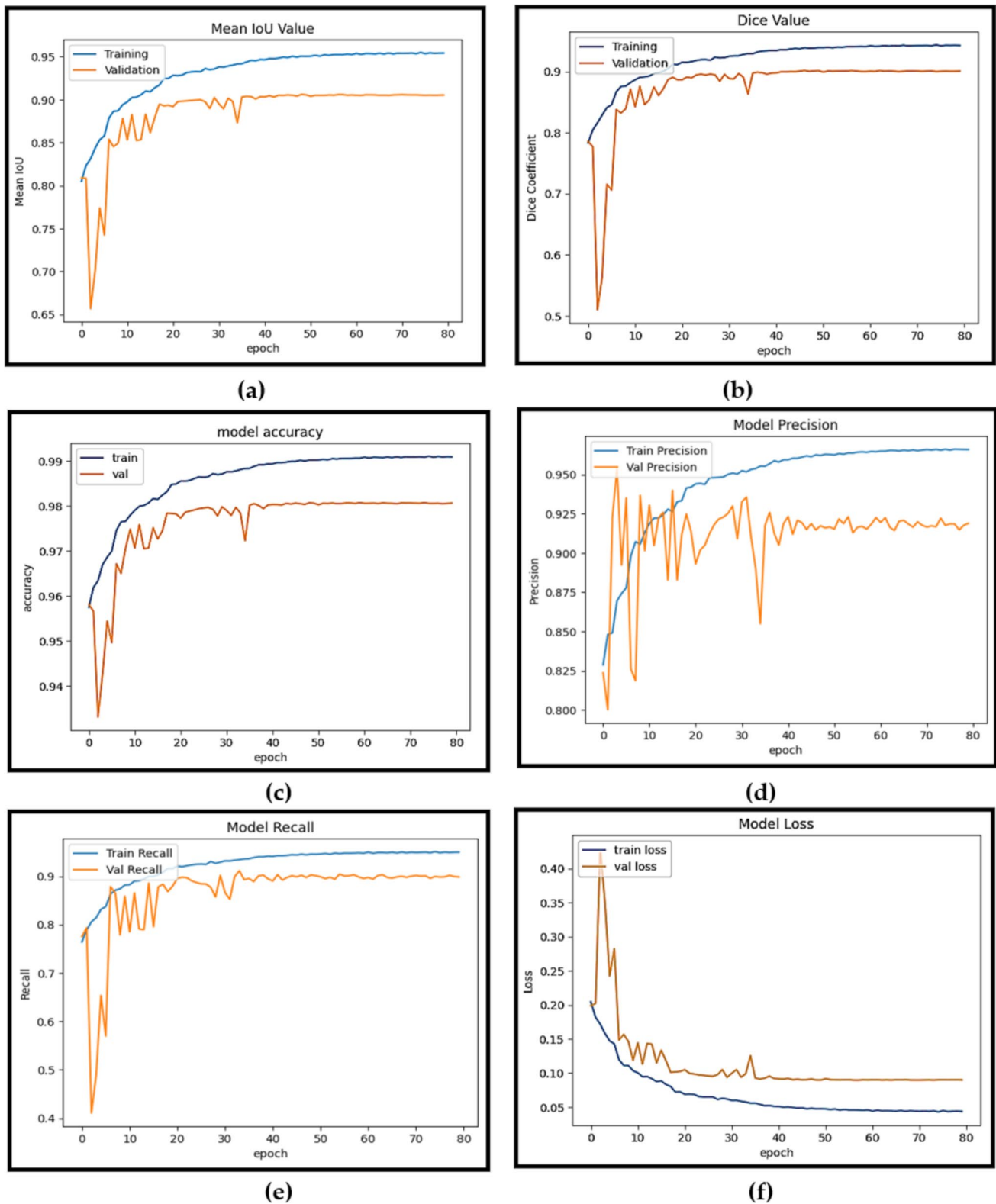


Fig. 10 Results of Proposed Model on Kvasir-SEG dataset, (a): Mean IoU, (b): Dice Coefficient, (c): Accuracy, (d): Precision, (e): Recall, (f): Loss

Table 5 Comparison of Proposed Model with state-of-the-art models on the CVC-ClinicDB Dataset

Sr. #	Model	Mean IoU	Dice	Recall	Precision
1	Polyp-PVT [67]	0.889	0.937	-	-
2	PlutoNet [68]	0.832	0.908	0.951	0.866
3	PRAPNet [62]	0.873	0.917	-	-
4	HSNet [69]	0.905	0.948	-	-
5	LAPFormer-L [70]	0.866	0.915	-	-
6	DS-TransUNet-L [60]	0.893	-	0.950	0.936
7	MSRF-Net [31]	0.904	0.942	0.956	0.930
8	GMSRF-Net [32]	0.888	0.932	0.937	0.930
9	FANet [61]	0.893	-	0.933	0.940
10	ColonFormer-S [65]	0.883	0.932	-	-
11	TGANet [63]	0.899	0.945	0.943	0.951
12	TA-Coreset [33]	0.887	-	-	-
13	GCCSwin-UNet [36]	0.846	0.912	-	-
14	FCB-SwinV2 [35]	0.826	0.900	0.943	0.869
15	HarDNet-CPS [34]	0.887	0.917	-	-
16	HarDNet with attention & multi-scale coding module [37]	0.892	0.931	-	-
17	Squeeze and Multi Context Attention Module [42]	0.74	0.82	0.88	0.74
18	PVT-Cascaded Attention Decoder [45]	0.899	0.943	-	-
19	ConvSegNet [38]	-	0.917	0.951	0.904
20	HMT-Unet [46]	0.909	0.952	-	-
21	SliceMamba [47]	0.897	0.946	-	-
22	Multi-Layer Dense Attention Decoder [49]	0.890	0.939	-	-
23	VMUnetV2 [50]	0.893	0.943	-	-
24	M3FPolySegNet [54]	0.850		0.901	0.906
25	BRAU-Net + + [55]	0.881	0.929	-	-
26	H-vmunet (H-VSS) [51]	-	0.908	-	-
27	LV-UNet [56]	0.851	0.91	-	-
28	FANetv2 [57]	0.903	0.948	0.951	0.949
29	Proposed Model	0.927	0.912	0.909	0.951

information. This feature is particularly advantageous for identifying subtle and small polyp regions. Our investigation demonstrates that the proposed encoder-decoder deep learning approach effectively segments polyps, including those that are flat or have low contrast and boundaries similar to the background, across various lighting conditions. The effectiveness of the proposed approach is reflected by the results that showed that the proposed approach achieved state-of-the-art mean IoU over Kvasir-SEG and CVC-ClinicDB datasets. The performance of the proposed approach suggests significant potential for enhancing colorectal polyp segmentation in colonoscopy images, which could have important implications for clinical practice. By improving the accuracy and efficiency of polyp detection, this approach may aid healthcare professionals in making more timely and informed decisions regarding patient diagnosis and treatment.

There are several limitations attached to the work done in this study. Firstly, the substantial size of the proposed model, comprising 39.9 million parameters, presents challenges for real-time applications and deployment on devices with limited computational resources. Secondly, the model requires evaluation on live-stream colonoscopy videos to verify its clinical applicability in real-world scenarios. Additionally, the sensitivity of the model to tissues resembling polyps and the presence of multiple small polyps may impact segmentation accuracy. To address these limitations, future work should focus on developing lightweight architecture or using model pruning and quantization techniques to reduce its size while maintaining high accuracy. Incorporating explainable AI techniques could also enhance model interpretability, which is crucial for clinical adoption. Furthermore, strategies to mitigate the effects of white light glare and other lighting variations should be explored, either through advanced pre-processing steps or by developing

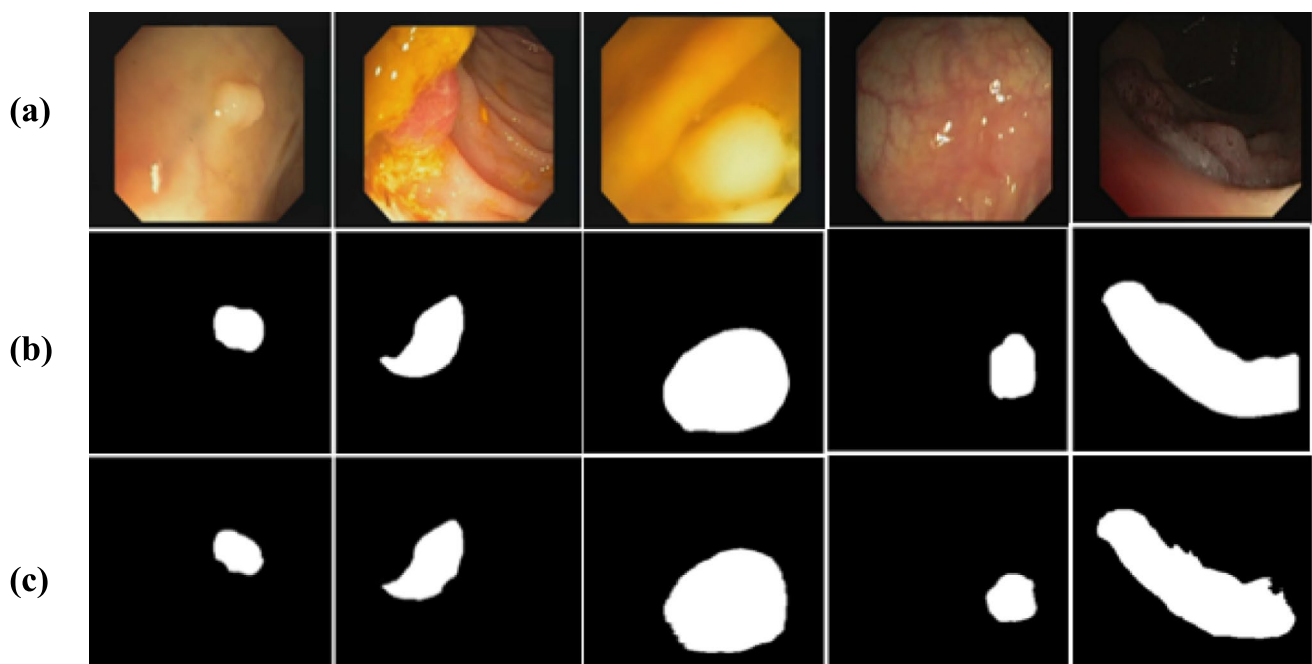


Fig. 11 Qualitative Results of Proposed Model on CVC-ClinicDB dataset (a): original image, (b): ground truth, (c): predicted mask

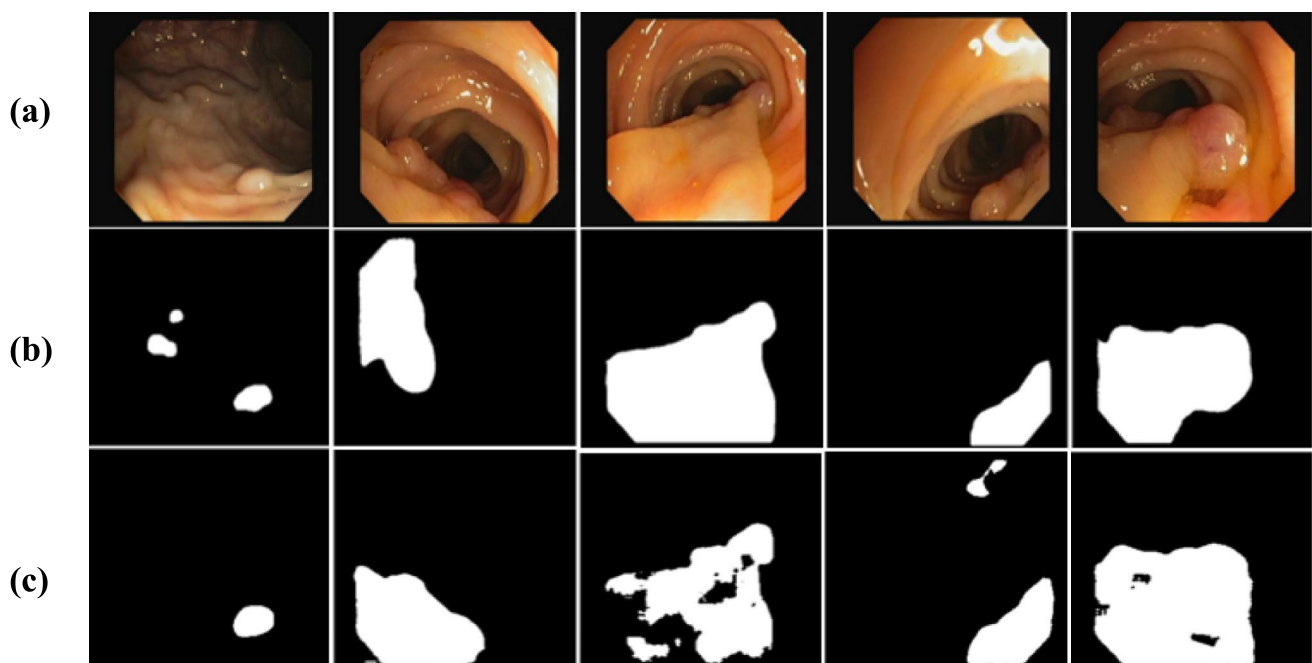


Fig. 12 Examples of challenging polyps cases in CVC-ClinicDB, (a): Original Image, (b): Ground Truth, (c): Predicted Mask

more robust algorithms. The proposed approach offers significant research and practical contributions, which will aid researchers and practitioners in the efficient detection of

polyps at an early stage. Continued refinement and validation of this model could potentially lead to improved clinical outcomes in colorectal cancer screening and prevention.

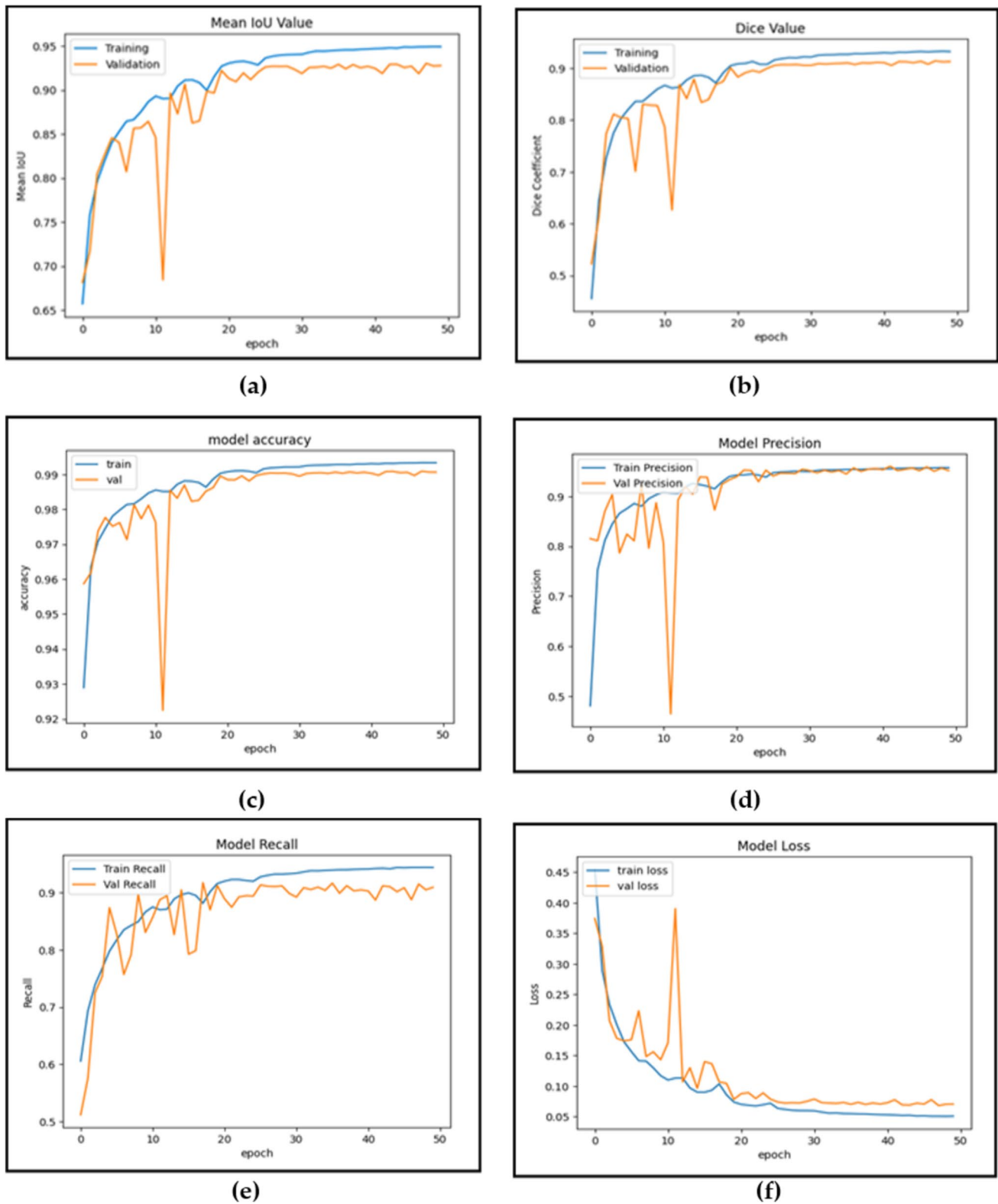


Fig. 13 Results of Proposed Model on CVC-ClinicDB dataset, (a): Mean IoU, (b): Dice Coefficient, (c): Accuracy, (d): Precision, (e): Recall, (f): Loss

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Data Availability The dataset used in this study is publicly available at <https://datasets.simula.no/kvasir-seg/> and <https://universe.roboflow.com/teste-mhyhc/cvc-clinicdb>

Declarations

Consent The publishers' "terms of use" are followed for the use of both published datasets.

Conflicts of Interest The authors declare no conflict of interest.

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